

Automatic Cycle Averaging for Denoising Approximately Periodic Spatiotemporal Signals

Weiguang Ding*, Eric Lin, Amanda Ribeiro, Marinko V. Sarunic, Glen F. Tibbits, and Mirza Faisal Beg

Abstract—Optical mapping has become a common tool for cardiovascular research, providing high resolution spatiotemporal data of cardiac action potential propagation. However, noise in cardiac optical mapping (COM) data hampers quantitative measurements and analyses. Spatial and temporal filters have been used to increase the signal-to-noise ratio (SNR) of COM data. Although these help reduce noise and increase SNR, they also lower the spatial and temporal resolution of the data, which is not desirable. This paper utilizes the approximate periodicity of COM data to perform denoising by averaging cycles. We consider the entire approximately periodic spatiotemporal (APST) COM data as a concatenation of random samples generated from a deterministic single cycle spatiotemporal signal. The image difference signal (IDS) was defined and calculated to provide “global” periodicity information. Parameters for cycle segmentation, scaling and alignment were estimated based on the IDS. Finally, these parameters were used to segment, align and average cycles from each individual signal, which produces a clean and denoised single cycle spatiotemporal signal. The novel, fully automated pipeline was validated both qualitatively and quantitatively on zebrafish heart optical mapping data.

Index Terms—Approximate periodicity, cycle averaging, optical mapping, spatiotemporal, time warping.

I. INTRODUCTION

OPTICAL imaging with voltage sensitive dyes, i.e., optical mapping, is a fundamental and common tool for cardiovascular function and disease research [1]–[5]. In cardiac optical mapping (COM), *ex vivo* animal hearts are immersed in solution and labelled with a fluorescent voltage-sensitive dye. Transmembrane voltage variation triggers fluorescence emission spectrum variation, which is captured by a color filtered

camera as movie data; each frame in it represents transmembrane voltage distribution at that moment. Also, the intensity changes of a specific location reflect voltage variation over time. These two aspects depict the spatiotemporal nature of optical mapping data. Recent comprehensive reviews of optical mapping techniques can be found in [1]–[3].

Raw cardiac optical mapping data are usually noisy and not directly suitable for measurements and analysis. Spatial and temporal filters are natural and effective ways for denoising COM data [6], and they are also used extensively in other spatiotemporal imaging modalities, for example functional magnetic resonance imaging (fMRI) [7]–[11]. However, spatial averaging mixes signals from different locations, which hampers classification of different cells [12], [13]. Similarly, temporal averaging lowers the temporal resolution, which affects measurements such as depolarization start, activation, peak moments, and action potential duration. Nonetheless, these and many other analyses [6] still require a certain level of signal-to-noise ratio (SNR), which necessitates the development of new denoising methods.

Harmonic analysis based methods have been used to denoise neural optical imaging signals [14]–[17], and also separate the non-periodic brain activity signal from periodic ballistocardiogram artifacts in electroencephalography (EEG) [18]. However, these techniques are mainly designed for analysis of periodic signals and may be suboptimal when cardiac activity deviates from strict periodicity. Principle component analysis (PCA) was used to denoise calcium imaging signals [19], but it is not suitable for cardiac optical mapping because the dominant principle components will ignore signal characteristics from small regions, for example pacemaker regions.

Cardiac optical mapping data are composed of many similar cardiac cycles, which enables another averaging dimension, cycle averaging. Previous work [20] averaged cardiac cycles stimulated by an external periodic artificial pacemaker. However, when it comes to intrinsically paced cardiac data from *ex vivo* hearts, the heart rate is affected by multiple factors. For example, the solution temperature might vary and the excised heart has very limited neural regulation (as no feedback from the central nervous system (CNS) exists). Under approximate periodicity (modelled and defined later in Section III-A), cycle averaging is not straightforward. To proceed, we need to solve two problems.

- How to separate the entire data into different cycles?
- How to align these slightly different cycles in order to average properly?

In this paper, we introduce a novel pipeline for denoising approximately periodic spatiotemporal (APST) data with cycle av-

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eraging. We first model different cycles of APST data as random samples generated from original single cycle data. Then, an image difference signal (IDS) is calculated for providing spatially global information for cycle separation and alignment. We suggest a heuristic cycle separation method specifically for COM data. Subsequently, temporal warping functions are acquired by aligning different cycles of IDS. Finally, these estimated parameters are used to segment, align and warp each individual COM signal. We use zebrafish heart optical mapping (ZHOM) data [12] for demonstration of this novel, automated pipeline, but the proposed method will work on other data with similar approximately periodic spatiotemporal properties.

II. EXPERIMENTAL SETUP

A. Data Acquisition

Zebrafish hearts were isolated and then soaked in the fluorescent voltage sensitive dye RH-237 [21] by immersion in 8 μM solution. The excitation-contraction uncoupler blebbistatin (10 μM) was used to inhibit contraction [22]. Excitation illumination was provided by a 200 mW 532 nm DPSS laser (Laserglow Technologies). Spectral filtering was provided by a 560-nm-long pass dichroic and a 710-nm-long pass emission filter (Omega Optical). Fluorescent images of the heart were acquired using a GE680 (Allied Vision Technologies) camera at a frame rate of 205 fps and resolution of 640×480 , followed by a 4×4 binning to reduce the resolution down to 160×120 . Masks for the atrium and ventricle were manually drawn on the acquired images.

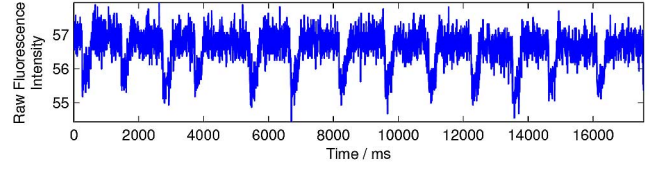
B. Preprocessing

The raw zebrafish heart optical mapping data has low SNR and low intensity resolution. Both spatial Gaussian filtering (with size [5, 5] and standard deviation 1) and temporal Gaussian filtering (with width 5 and standard deviation 1) were performed on the original data. The signal drift was represented by fitting the data to a quadratic curve [6], which was then subtracted from the original signal. Finally, the signal values were inverted and linearly scaled to [0, 1].

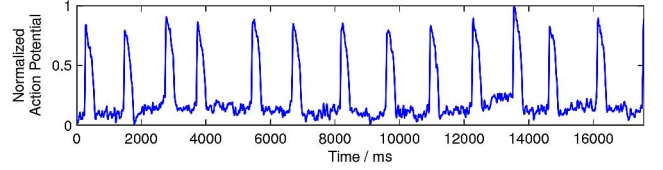
Fig. 1 shows an example of preprocessed ZHOM data. The raw voltage induced fluorescence intensity from a single pixel of ZHOM data is presented in (a). Preprocessed normalized action potentials are shown in (b). Visualization of action potential propagation on a zebrafish heart is presented in (c).

C. Data Properties

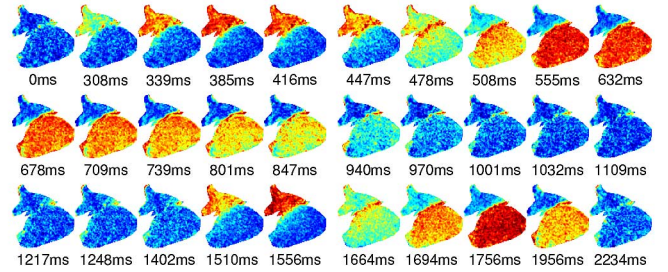
The data shown in Fig. 1 have gone through significant amount of spatial and temporal averaging and they cannot afford further loss on spatial and temporal resolution. However, the data are still noisy, and the noise corrupts all subsequent analyses, as discussed in Section I. For example, diastolic depolarization, a hallmark of the sinoatrial node [23], is measured from the action potential signal while the heart is at rest. These measurements are particularly sensitive to noise because of their small absolute signal values. Therefore, cycle averaging is needed for further denoising in this case.



(a) Raw voltage signal from single pixel of ZHOM data



(b) Voltage signal after spatial and temporal averaging



(c) Visualization of zebrafish heart action potential propagation

Fig. 1. Example of approximately periodic spatiotemporal data. Panel (a) shows a raw fluorescence intensity signal from a single pixel of ZHOM data. Panel (b) shows normalized action potentials after preprocessing. Panel (c) visualizes the action potential propagation of a zebrafish heart. These pseudo color maps represent voltage values at different pixels. Each pseudo color map reflects the action potential distribution at a specific time point. Here, red means high voltage (depolarization) and blue low voltage (hyperpolarization).

To ensure approximate periodicity, we ensured that hearts used for cycle averaging do not have second-degree atrioventricular block [24], which means atrial impulses would fail to conduct to the ventricle. This would break the approximate periodicity. But we allow the heart to have first-degree atrioventricular block [25] or other problems that will not cause the ventricle to skip cycles. For example, different cardiac cycles might have different durations. This is further discussed in Section VI-B4.

III. MODELLING

We use $f(x, y, t)$ to represent approximately periodic spatiotemporal signals, where (x, y) and t represent spatial and temporal locations respectively. As a function of time, $f^{x,y}(t)$ is used to represent a 1-D signal at a specific location (x, y) . As $f^{x,y}(t)$ is “approximately periodic,” it can be represented by its cycles in a piecewise manner

$$f^{x,y}(t) = \begin{cases} f_1^{x,y}(t), & \text{if } 0 \leq t \leq \tau_1^{x,y} \\ f_2^{x,y}(t - \tau_1^{x,y}), & \text{if } \tau_1^{x,y} < t \leq \tau_2^{x,y} \\ \dots & \dots \\ f_i^{x,y}(t - \tau_{i-1}^{x,y}), & \text{if } \tau_{i-1}^{x,y} < t \leq \tau_i^{x,y} \\ \dots & \dots \\ f_n^{x,y}(t - \tau_{n-1}^{x,y}), & \text{if } \tau_{n-1}^{x,y} < t \leq \tau_n^{x,y} \end{cases} \quad (1)$$

where $i = 1, 2, \dots, n$ represents the indexes of different cycles; $\tau_i^{x,y}$ stands for the stopping time of cycle i (inclusive) and

is also the start of cycle $i + 1$ (exclusive); $f_i^{x,y}(t)$ represents cycle i of the full signal $f^{x,y}(t)$. Thus, $f_i^{x,y}(t)$ is defined on interval $(0, \tau_i^{x,y} - \tau_{i-1}^{x,y}]$, with an exception of $f_1^{x,y}(t)$ defined on $[0, \tau_1^{x,y}]$.

A. Cycle Generation

Inspired from previous work on signal estimation and alignment [26], we assume that individual cycles $f_i^{x,y}(t)$'s are random samples of a single cycle signal $g^{x,y}(t)$

$$f_i^{x,y}(t) = g^{x,y}(\gamma_i^{x,y}(a_i^{x,y}t)) + \varepsilon_i^{x,y}(t), \quad i = 1, 2, \dots, n \quad (2)$$

where $g^{x,y}(t)$ is a real valued function defined on the normalized temporal interval $[0, 1]$; $\gamma_i^{x,y} : [0, 1] \rightarrow [0, 1]$ is a 1-D diffeomorphism with $\gamma_i^{x,y}(0) = 0$ and $\gamma_i^{x,y}(1) = 1$; $a_i^{x,y} \in \mathbb{R}$ is a temporal scaling factor; and $\varepsilon_i^{x,y}(t)$ is a zero-mean additive white Gaussian noise term, which is uncorrelated with respect to either t , x , y , or i . This gives the following assumption.

Assumption 1: (Random cycle generation) Individual signal cycles of an approximately periodic signal are random samples of an unknown deterministic signal under stochastic temporal scaling, temporal warping, and additive noise.

In other words, we defined ‘‘Approximate periodicity.’’ A signal is approximately periodic if it is a concatenation of random samples of a deterministic signal under stochastic temporal scaling, temporal warping and additive noise.

B. Parameter Sharing

For practical APST data, temporal signals, $f^{x,y}(t)$'s, from different locations, (x, y) 's, usually keep spatial correlation with each other. For example in COM data, each action potential signal usually has fixed phase correlations with its neighboring signals, as the mechanism of action potential generation and propagation is shared. Based on this observation, the second assumption about spatial uniformity of warping and scaling is:

Assumption 2: (Parameter sharing) Signals from different locations share the same cycle separation time. Furthermore, for any given cycle, signals from different locations share the same random warping function and temporal scaling factors.

Mathematically, Assumption 2 means that $\tau_i^{x,y}$, $\gamma_i^{x,y}(t)$, and $a_i^{x,y}$ are not functions of (x, y) . Therefore, they can be written simply as τ_i , $\gamma_i(t)$ and a_i . Hence, (1) is rewritten as

$$f^{x,y}(t) = \begin{cases} f_1^{x,y}(t), & \text{if } 0 \leq t \leq \tau_1 \\ f_2^{x,y}(t - \tau_1), & \text{if } \tau_1 < t \leq \tau_2 \\ \dots & \dots \\ f_i^{x,y}(t - \tau_{i-1}), & \text{if } \tau_{i-1} < t \leq \tau_i \\ \dots & \dots \\ f_n^{x,y}(t - \tau_{n-1}), & \text{if } \tau_{n-1} < t \leq \tau_n \end{cases} \quad (3)$$

and (2) is rewritten as

$$f_i^{x,y}(t) = g^{x,y}(\gamma_i(a_i t)) + \varepsilon_i^{x,y}(t), \quad i = 1, 2, \dots, n. \quad (4)$$

C. Illustration of Signal Generation Process

Fig. 2 shows an example of generating an approximately periodic signal from a single original one cycle signal. The original signal $g^{x,y}(t)$ is shown in (a); time warping functions $\gamma_i(t)$'s are

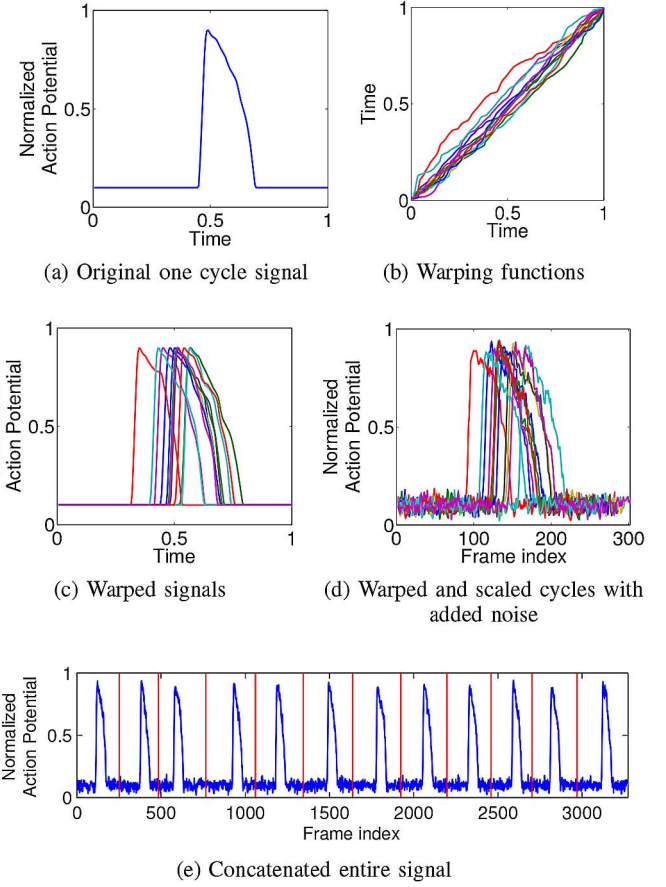


Fig. 2. Illustration of signal generation. Panel (a) shows the original one cycle signal. Panel (b) shows temporal warping functions. Panel (c) shows the temporally warped signals. Panel (d) shows the warped, scaled and noised signals. Panel (e) shows the entire signal concatenated from cycles in (d). Red vertical lines in (e) are separators for different individual cycles.

contained in (b). They are used to warp the original signals to acquire the deformed $g^{x,y}(\gamma_i^{x,y}(t))$'s as shown in (c). The warped, scaled, and noise-added signals $g^{x,y}(\gamma_i^{x,y}(a_i^{x,y}t)) + \varepsilon_i^{x,y}(t)$'s are shown in (d). After concatenating these deformed signals, we generated an approximately periodic signal as shown in (e).

IV. METHODS

Following the modelling of signal generation, we consider the procedure of cycle averaging process as the estimation of the original signal $g^{x,y}(t)$ from every $f^{x,y}(t)$ on different (x, y) locations. If the parameters τ_i , a_i , and $\gamma_i(t)$ are known, the following equations can be used for estimation:

$$f_i^{x,y}(t) = f^{x,y}(t + \tau_{i-1}), \quad t \in [0, \tau_i - \tau_{i-1}] \quad (5)$$

$$\hat{g}^{x,y}(t) = \frac{\sum_{i=1}^n f_i^{x,y}(\frac{\gamma_i^{-1}(t)}{a_i})}{n} \quad (6)$$

where $\hat{g}^{x,y}(t)$ represents the estimation of $g^{x,y}(t)$. Equations (5) and (6) constitute the reverse procedure of signal generation represented by Equations (3) and (4). Performing this procedure requires the following steps.

- 1) Estimate τ_i 's to separate the entire $f^{x,y}(t)$ into different single cycle signals $f_i^{x,y}(t)$'s.

- 2) Estimate the temporal scaling factor a_i and warping function $\gamma_i(t)$ to deform each $f_i^{x,y}(t)$ to acquire a single noisy cycle estimation $f_i^{x,y}(\gamma_i^{-1}(t)/a_i)$.
- 3) Average over these single estimations to get the final estimation $\hat{g}^{x,y}(t)$.

A. Image Difference Signal

Estimations of τ_i , $\gamma_i(t)$ and a_i are “spatially global” task for the entire APST data. In other words, we need to make consistent estimations for different spatial locations. Therefore, to ensure this consistency and make full use of the data, we proposed to use the “image difference signal” (IDS) for estimation of the cycle separation, warping and scaling parameters. The IDS $\rho(t)$ is calculated by the following equation:

$$\rho(t) = \|I_t - I_{\text{ref}}\|^2 = \sum_{x,y} (I_t(x,y) - I_{\text{ref}}(x,y))^2 \quad (7)$$

where I_t is an image at time t which has $I_t(x,y) = f^{x,y}(t)$'s as its pixels, and I_{ref} is the reference image which has $I_{\text{ref}}(x,y)$'s as its pixels.

According to (3) and (7), $\rho(t)$ can also be divided into different cycles $\rho_i(t)$'s in a similar way as $f^{x,y}(t)$

$$\rho(t) = \begin{cases} \rho_1(t), & \text{if } 0 \leq t \leq \tau_1 \\ \rho_2(t - \tau_1), & \text{if } \tau_1 < t \leq \tau_2 \\ \dots & \dots \\ \rho_i(t - \tau_{i-1}), & \text{if } \tau_{i-1} < t \leq \tau_i \\ \dots & \dots \\ \rho_n(t - \tau_{n-1}), & \text{if } \tau_{n-1} < t \leq \tau_n. \end{cases} \quad (8)$$

Furthermore, each $\rho_i(t)$ can be expressed in the same way as $f_i^{x,y}(t)$ as shown in (4). That is, for each $\rho_i(t)$, there exists a function $\eta(t)$ defined on $[0, 1]$ such that

$$\rho_i(t) = \eta(\gamma_i(a_i t)) + \mathcal{E}_i(t), \quad i = 1, 2, \dots, n. \quad (9)$$

Here, $\eta(t)$ is the “original one cycle” image difference signal, which is similar to $g^{x,y}(t)$. $\mathcal{E}_i(t)$ is a zero mean error term with $\text{Var}(\mathcal{E}_i(t)) = 2 \sum_{x,y} \sigma^4 + 4 \sum_{x,y} \sigma^2 (g^{x,y}(\gamma_i(a_i t)) - I_{\text{ref}}(x,y))^2$. The detailed derivation and analysis of (9) is discussed in Appendix A.

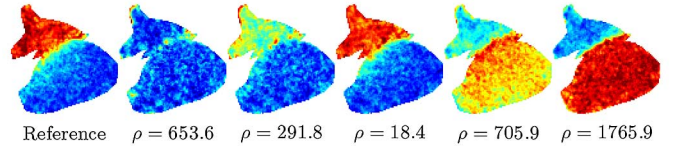
Following (8) and (9), the image difference signal $\rho(t)$ can be used to estimate τ_i , $\gamma_i(t)$, and a_i . As desired, these estimations from $\rho(t)$ are consistent across different spatial locations and utilize information from the entire spatiotemporal data.

The choice of reference image should reflect the difference between images in $\rho(t)$. Specifically for the zebrafish heart optical mapping data used in this paper, we select I_{t_0} as the reference image where t_0 maximizes the average signal value inside the atrial region of interest (ROI)

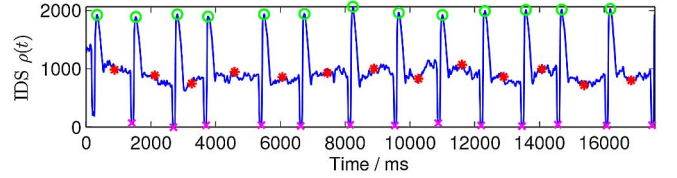
$$t_0 = \arg \max_t \sum_{(x,y) \in A} I_t(x,y) \quad (10)$$

where A is a set containing all the pixel coordinates in the atrium.

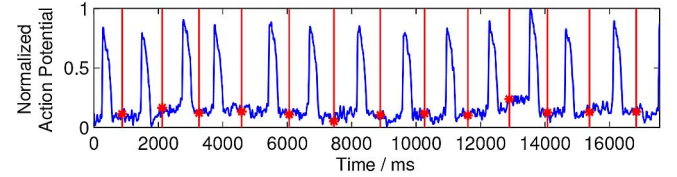
In this case, peaks in $\rho(t)$ correspond to the maximal ventricular depolarization events, and valleys correspond to maximal atrial depolarization events in different cycles.



(a) Reference image I_{ref} and different $I(t)$'s with corresponding ρ 's



(b) Image signal difference $\rho(t)$ with peaks, valleys and midpoints labelled



(c) Specific signal with each cycle identified and separated

Fig. 3. Illustration of image difference and cycle separation results. Panel (a) shows the chosen reference images, different representative images and their corresponding image difference values. Panel (b) shows the IDS with its peaks (green o's), valleys (magenta x's) and midpoints (red *'s) labelled. Panel (c) shows an example of entire signal with its cycle separated.

B. Cycle Separators and Scaling Factors

Empirically in our zebrafish heart optical mapping data, peaks in $\rho(t)$ satisfy.

- 1) Peak values are above $0.7 \max_t \rho(t)$.
- 2) Distance between adjacent peaks is larger than half of the average period.

Here the average period was roughly estimated as the $\rho(t)$ length divided by number of cycles in $\rho(t)$. The number of cycles was estimated by half of the number of zero crossings in the signal $\rho(t) - 0.7 \max_t \rho(t)$

$$\text{AveragePeriod} = \frac{2 \text{Length}(\rho(t))}{\# \text{ZeroCrossings}(\rho(t) - 0.7 \max_t \rho(t))}.$$

Similarly, valleys in $\rho(t)$ are peaks in $\rho'(t) = \max_t \rho(t) - \rho(t)$, which also satisfy the above criteria. We then utilized MATLAB [27] inbuilt function *findpeaks* to find peaks and valleys with the stated criteria.

With the peaks and valleys detected, determining the separation times τ_i 's is an arbitrary choice. Specifically for ZHOM data, we chose midpoints between peaks and valleys in $\rho(t)$ as estimated cycle separators $\hat{\tau}_i$'s.

Estimation of a_i 's is straightforward given $\hat{\tau}_i$'s, as $\gamma_i(t)$'s are defined on interval $[0, 1]$

$$\hat{a}_i = \hat{\tau}_i - \hat{\tau}_{i-1}. \quad (11)$$

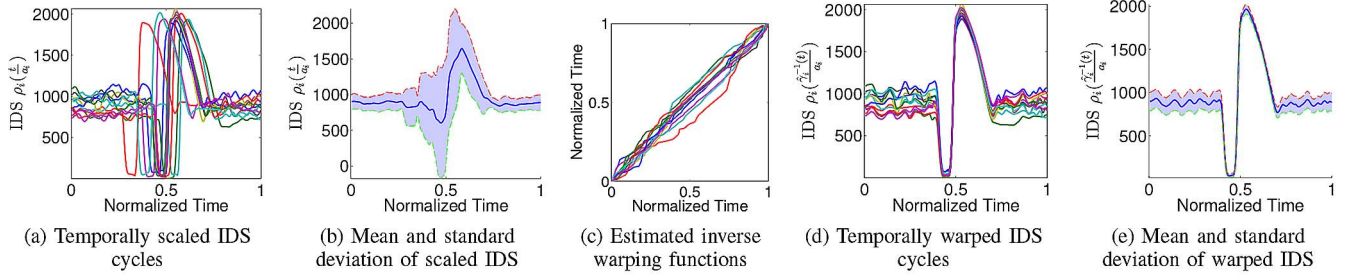


Fig. 4. Estimation of warping function. Panel (a) shows the scaled but unwrapped individual IDS cycles $\rho_i(t/a_i)$'s. Panel (b) shows the estimated inverse warping functions $\hat{\gamma}_i^{-1}(t)$'s using Kurtek's method. Panel (c) shows the scaled and warped individual IDS cycles $\rho_i(\hat{\gamma}_i^{-1}(t)/a_i)$'s. Panel (d) shows the mean and standard deviation of averaged scaled but unwrapped individual IDS cycles. Panel (e) shows the mean and standard deviation of averaged scaled and warped individual IDS cycles. Shaded areas in (d) and (e) are within 1 standard deviation of signal values at that given time.

So far, as τ_i 's and a_i 's are estimated, the entire signal $f^{x,y}(t)$ has been separated into different cycles and stretched to interval $[0, 1]$.

C. Warping Function Estimation

Before the final averaging step, we need to estimate $\gamma_i(t)$ in order to align these randomly deformed individual cycles. We adopted the signal estimation and nonlinear signal alignment method proposed by Kurtek *et al.* [26]. This method is able to estimate the unknown signal $\eta(t)$ and the random time warping functions $\gamma_i(t)$ from $\rho(t)$ simultaneously. This method first constructs an equivalence relation between signals using warping group action. Next, the Karcher mean is used to estimate the equivalence class of the unknown original signal on the quotient space of equivalence classes. The unknown original signal is treated as the center of the estimated equivalence class. Finally, the original signal and the warping functions are estimated simultaneously in an iterative manner. For a detailed explanation of this method, please refer to [26], [28].

D. Cycle Averaging for Individual Signals

So far, we have acquired all of the parameters: τ_i , $\gamma_i(t)$, and a_i . We then estimated the original signal $g_{x,y}(t)$ for each location (x, y) by performing (5) and (6).

V. RESULTS

A. Qualitative

Fig. 3 shows the results of image difference signal calculation and cycle separation. The leftmost panel in (a) is the reference image I_{ref} chosen as described previously. The remaining images in (a) are different I_t 's extracted from the spatiotemporal data. When I_t is similar to I_{ref} , the ρ has small value and vice versa. Fig. 3(b) shows an example of the entire image difference signal. For our specific application of processing zebrafish heart optical mapping data, we chose to use the midpoints between peaks and valleys in $\rho(t)$ as the cycle separators. The green o's represent detected peaks, magenta x's represent detected valleys, and red *'s are the midpoints used as separators. Fig. 3(c) shows an example of entire signal from a single spatial location of the data. It is separated into different cycles using the red *'s acquired from the image difference signal $\rho(t)$.

Fig. 4 shows the results related to estimation of warping functions. Individual IDS cycles segmented from the entire IDS

shown in Fig. 3(b) are shown in (a). These cycles are temporally scaled to interval $[0, 1]$ and mathematically expressed as $\rho_i(t/a_i)$. The mean and standard deviation of averaged scaled individual IDS cycles before warping are shown in (b). The estimated inverse warping functions $\hat{\gamma}_i^{-1}(t)$'s using Kurtek's method are presented in (c). Scaled and warped individual IDS cycles $\rho_i(\hat{\gamma}_i^{-1}(t)/a_i)$'s are shown in (d). The mean and standard deviation of averaged scaled individual IDS cycles after warping are shown in (e). Comparing (b) and (e), the average of warped one cycle IDS keeps the original shape and has smaller standard deviation. The average of unwrapped one cycle IDS as shown in (b) in comparison loses the original signal shape and has a larger standard deviation.

Fig. 5 shows the cycle averaging procedure on individual signals. Following the cycle segmentation shown in Fig. 3(c), Fig. 5(a) contains the scaled but unwrapped individual signal cycles $f_i(t/a_i)$'s defined on interval $[0, 1]$. These cycles are then warped to become $f_i(\hat{\gamma}_i^{-1}(t)/a_i)$'s as shown in (b). Subfigures (c) and (d) compare the mean and standard deviation between warped and unwrapped single cycle signals. The visualization of action potential propagation after cycle averaging is shown in (e). By performing cycle averaging, the action potential propagation has a significant increase in SNR. Subfigure (f) compares propagation visualizations before and after cycle averaging side by side. Before cycle averaging, the propagation appears granular and noisy; after cycle averaging, it shows a smooth color coded propagation wavefront.

Fig. 7 shows more representative examples of estimated single cycle signals taken from different hearts. For each of these examples, a full length signal with the cycle separators labelled is shown in (a). Individual cycles before and after warping are presented in (b) and (d), respectively. The mean and standard deviation of individual cycles before and after warping are shown in (c) and (e), respectively. Sub-panel (f) compares propagation visualization before and after cycle averaging.

B. Quantitative

We acquired cardiac optical mapping data from five different zebrafish hearts under 28°C for quantitative analysis. Among these five recordings, four of them contain 40 cycles which are used to evaluate cycle duration variability, SNR, and run-time. The fifth heart contains 320 cycles, and is used to evaluate the SNR increase with respect to a large number of cycles.

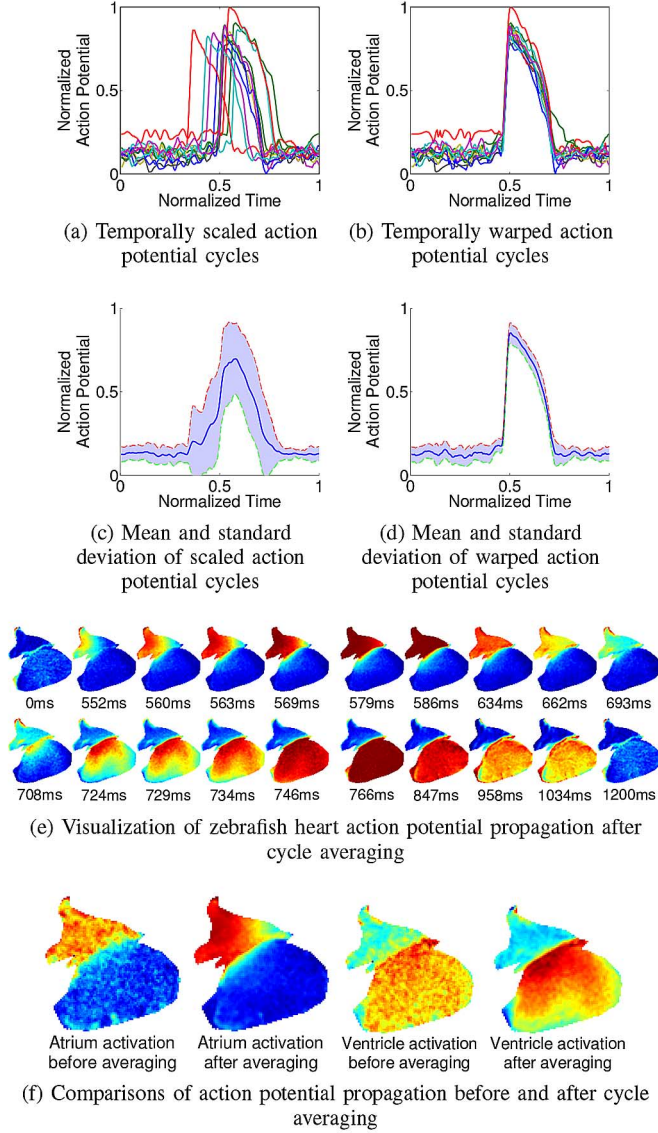


Fig. 5. Cycle averaging for denoising individual signals. Panel (a) shows the scaled but unwarped individual signal cycles $f_i(t/a_i)$'s; Panel (b) shows the scaled and warped individual signal cycles $f_i(\hat{\gamma}_i^{-1}(t)/a_i)$'s; Panel (c) shows the mean and standard deviation of averaged scaled but unwarped individual signal cycles; Panel (d) shows the mean and standard deviation of averaged scaled and warped individual signal cycles. Shaded areas in (c) and (d) are within 1 standard deviation of signal values at that given time. Panel (e) visualizes the action potential propagation after cycle averaging. Panel (f) compares both atrium and ventricle action potential propagation before and after cycle averaging.

1) *Cycle Duration Variability*: Difficulties of averaging across cycles arise from cycle variability. Here, we consider the standard deviation of cycle durations from one full length signal as a quantitative measure of cycle variability. The averaged standard deviation of cycle duration is 1.3% of the mean cycle duration. The averaged peak-to-peak difference (difference between maximum and minimum) of cycle duration is 5.0% of mean cycle duration. In one of these hearts, the peak-to-peak difference reached 9.2% of mean cycle duration.

2) *Signal-To-Noise Ratio*: We define the SNR as the quotient of signal amplitude divided by the peak-to-peak noise in the resting period [29], which is considered as the normalized

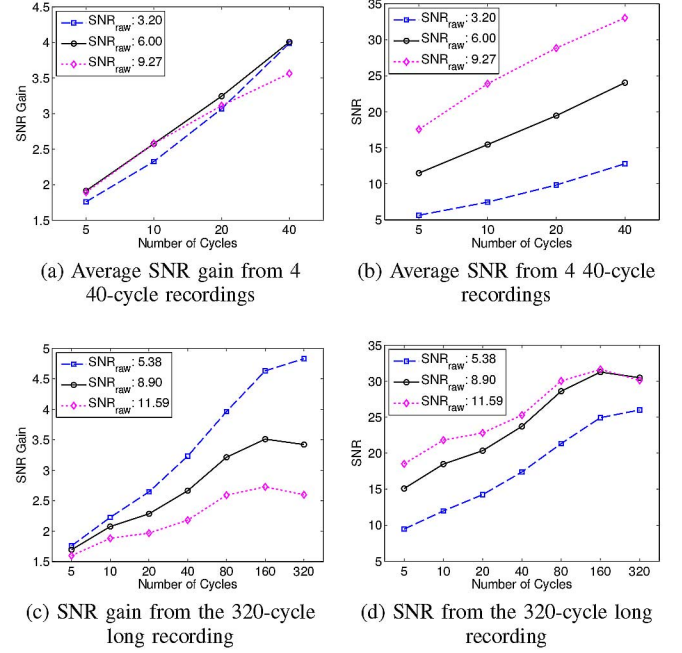


Fig. 6. SNR gain and SNR as functions of number of cycles. Different plots within each individual panel are preprocessed with spatial and temporal filters of different sizes, which lead to different raw SNRs (shown as legends). Both SNR gain and SNR are defined as ratios, hence do not have associated units. See Section V-B2 for details.

temporal interval $[0.05, 0.3]$ for warped individual signal cycles [Fig. 5(c) and (d)]. Mathematically, this is represented by the formula

$$\text{SNR} = \frac{\max_{t \in [0,1]} \hat{g}^{x,y}(t) - \min_{t \in [0,1]} \hat{g}^{x,y}(t)}{\max_{t \in [0.05,0.3]} \hat{g}^{x,y}(t) - \min_{t \in [0.05,0.3]} \hat{g}^{x,y}(t)}. \quad (12)$$

For warped raw cycles, we acquired the raw SNR, SNR_{raw} , by first calculating the SNR for each single cycle, and then averaging them over cycles, pixels and hearts. Similarly, we calculated the averaged cycle SNR, SNR_{avg} , for the averaged signal by first calculating the SNR for each single cycle, and then averaging them over pixels and hearts. We also define SNR gain, gain_{SNR} , as the ratio of the raw SNR to the averaged cycle SNR

$$\text{gain}_{\text{SNR}} = \frac{\text{SNR}_{\text{avg}}}{\text{SNR}_{\text{raw}}}. \quad (13)$$

Fig. 6 plots the SNR gain and the SNR as functions of the number of cycles.

To see the performance of the algorithm under different SNR_{raw} 's, we used three sets of spatial and temporal Gaussian filter parameters: $[3, 3]$ spatial, width 3 temporal and standard deviation 0.6; $[5, 5]$ spatial, width 5 temporal and standard deviation 1; $[7, 7]$ spatial, width 7 temporal and standard deviation 1.6. Fig. 6(a) and (b) show the averaged results from four different hearts, each with a 40-cycle recording. Fig. 6(c) and (d) show results from one 320-cycle recording from the fifth heart. In Fig. 6(a) and (b), when the number of cycles is relatively small, both the SNR gain and the SNR are roughly linear to the logarithm of the number of cycles. In Fig. 6(c), when the number of cycles becomes larger, the SNR gain

TABLE I
RUN-TIME MEASUREMENTS

	Total	Segmentation Estimation	Warping Estimation	Segmentation-Interpolation	Warping Execution	Cycle Averaging
Time (s)	82.5	1.1	38.6	21.3	21.4	0.2

(a) Run-time of each component under our common experiment settings

Frames per Cycle	Number of Cycles			
	5	10	20	40
100	1.2	2.6	5.5	10.6
200	4.7	8.8	19.7	38.6
400	17.4	34.7	69.3	153.2
800	69.6	142.0	278.3	567.6

(b) Warping estimation

Frames per Cycle	Number of Cycles			
	5	10	20	40
100	2.9	5.3	10.3	20.8
200	3.0	5.5	10.5	21.3
400	3.0	5.8	11.2	22.6
800	3.2	6.3	12.5	25.4

(c) Cycle segmentation-interpolation

Frames per Cycle	Number of Cycles			
	5	10	20	40
100	2.5	4.6	9.2	18.9
200	2.8	5.4	10.6	21.4
400	3.3	6.5	12.7	26.1
800	4.3	8.5	17.2	35.9

(d) Warping execution

tends to saturate and stops increasing. With respect to different SNR_{raw} values, data with lower SNR_{raw} have larger SNR gain through cycle averaging. Accordingly in Fig. 6(d), the SNR_{avg} 's given different SNR_{raw} 's tend to converge to the same value, which means cycle averaging is able to provide a signal with the same SNR_{avg} with less spatial and temporal averaging. Limitations of averaging large number of cycles are discussed in Section VI-B5.

3) *Run-Time*: The working environment was a laptop with eight-thread 2.4-GHz CPU (The processor name is Intel Core i7-4700MQ. Our programs only utilized one thread.) and 16-GB memory running 64-bit Windows 7 and MATLAB 2012b.

We timed these experiments [shown in Table I(a)] under our common settings: 40 raw cycles (about 3600 original frames), image size 160×120 and warped cycle with 200 frames. According to Table I(a), segmentation estimation and cycle averaging are very fast; warping estimation, segmentation-interpolation, and warping execution consume most of the run-time. Therefore, we timed these time-consuming stages separately with respect to number of cycles, n_c and number of frames per cycle, n_f . Results are shown in Table I(b), Tables I(c), and I(d). Based on these empirical results, the complexity of warping estimation is approximately $O(n_c n_f^2)$, which is linearly proportional to the number of cycles and quadratic to number of frames per cycle; complexities of cycle segmentation-interpolation and warping execution are both approximately $O(n_c)$ and sub-linear with respect to n_f .

Segmentation estimation, cycle segmentation-interpolation and cycle averaging can compose a linear cycle averaging pipeline by themselves; warping estimation and execution are essential nonlinear stages of the complete proposed pipeline. When n_c and n_f are small, the run-times of linear and nonlinear pipelines are within the same order of magnitude; but when n_c and n_f are large, a nonlinear pipeline consumes significantly more time.

VI. CONCLUSION AND DISCUSSION

A. Conclusion

This paper describes a novel pipeline for denoising approximately periodic spatiotemporal (APST) data. APST data were modelled as concatenation of independently generated random samples from single cycle spatiotemporal data. Based on this model, we used the image difference signal to provide “global”

information, which was later used for estimating segmentation, scaling and alignment parameters. These estimated parameters were then used to segment, align and finally average individual signal cycles. The inputs of this pipeline are noisy APST data containing multiple cycles. The outputs are clean and denoised single cycle spatiotemporal data, which share the same spatial resolution as the input data. A side-by-side comparison of images before and after averaging [shown in Fig. 5(f), and Fig. 7(f)] shows that the proposed cycle averaging method enhances both small and large scale spatial features of the action potential propagation [Fig. 5(e)] that were not evident in the spatial and temporal filtered results [Fig. 1(c)]. By revealing these finer features, the resulting high fidelity action potential may lead to discovery of novel underlying patterns and mechanisms that are not evident in the standard action potential cycle.

We conclude that the proposed method shows significant performance for denoising APST data in three aspects.

- The parameters estimated from IDS enable each individual signal to be temporally aligned properly.
- The cycle-averaged signals are visually smoother than original individual cycles and have higher SNR.
- The denoised action potential propagation visualization images are less granular and clearly show the propagation wavefront shape which can be hardly seen in the original data.

B. Discussion

1) *Comparisons With Previous Methods*: The cycle alignment algorithm used here is the same as the one used in the previous automatic signal segmentation and alignment method [28]. However, this method is not directly usable on cardiac optical mapping data for three reasons.

- 1) If the segmentation, alignment, and warping are performed for individual signals, the averaged signals will be no longer temporally consistent with each other.
- 2) A single signal is usually noisy and cannot be used for cycle segmentation and alignment directly.
- 3) This method still requires manual initialization in cycle segmentation, which is not suitable for high throughput analysis.

In contrast, our method utilizes the image difference signal to estimate cycle segmentation and alignment parameters, which

- 1) is representative for the entire APST data;
- 2) avoids dealing with individual noisy signals;

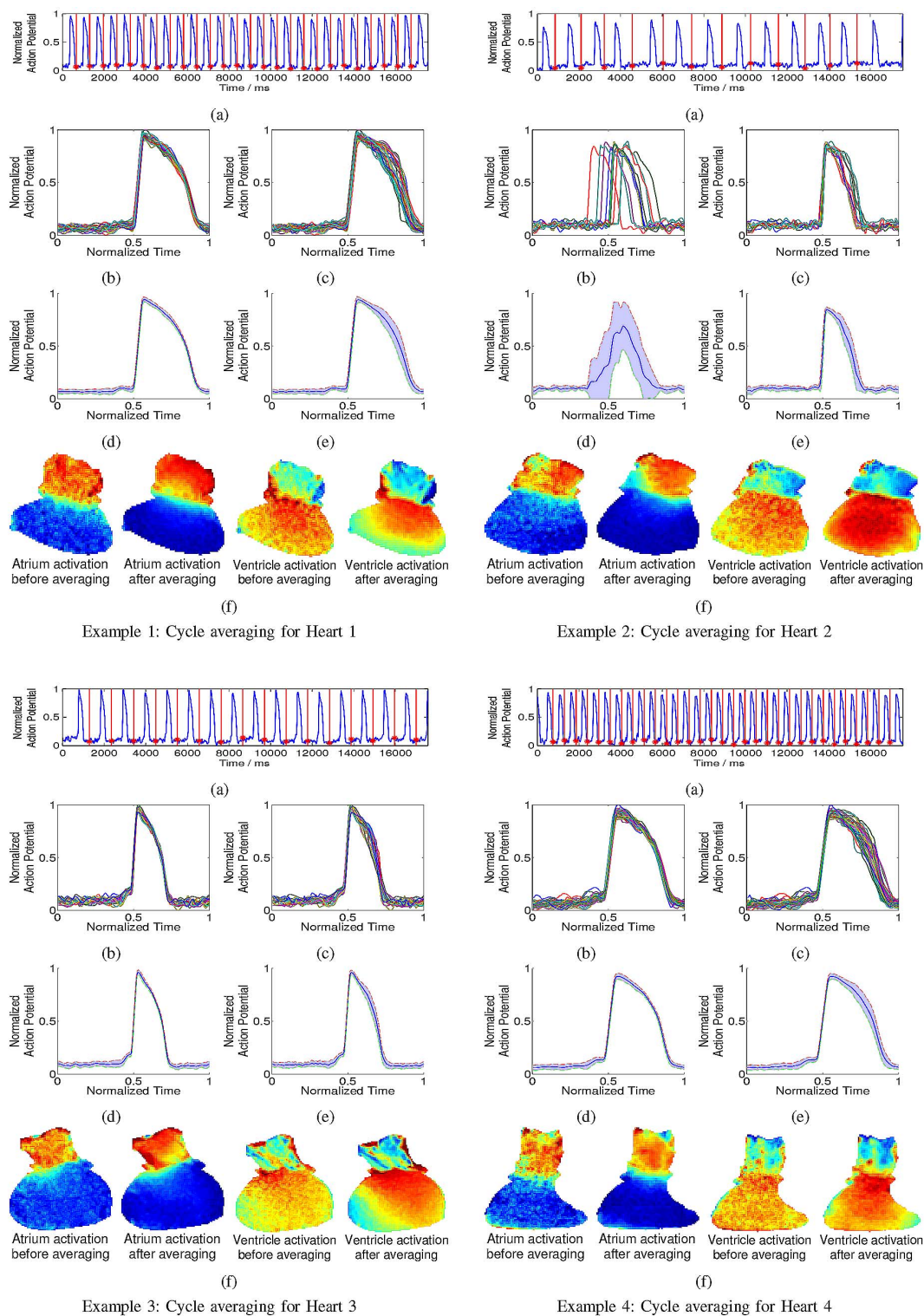


Fig. 7. More cycle averaging examples. For each of these examples in different panels: sub-panel (a) shows a full length signal with cycle separator labelled; sub-panel (b) shows the individual cycles before warping; sub-panel (c) shows the individual cycles after warping; sub-panel (d) shows the mean and standard deviation of individual cycles before warping; sub-panel (e) shows the mean and standard deviation of individual cycles after warping; sub-panel (f) compares both atrium and ventricle action potential propagation before and after cycle averaging.

3) is fully automatic.

The previous cycle averaging method for COM data [20] relies on an external input to segment cycles. Furthermore, it can only be used when the cardiac tissue excitation is induced by artificial stimulation. Harmonic analysis based methods [14]–[18]

are usually used to analyze periodic signals and have potential extensions to non-periodic signals. In comparison, our method is designed for the assumed approximate periodicity and therefore will likely work better on data with similar properties.

2) *Potential Variations of the Proposed Pipeline*: This paper proposes a general segmentation and alignment pipeline, which is not constrained on the specific algorithms used in each step. The image difference signal can be replaced with any other signal that is able to reflect global variation of the APST data, such as image correlation. As stated in Section IV, the reference image selection and cycle segmentation methods are also not fixed and can be tailored for specific applications. For the reference image, we also have the option to use multiple reference images simultaneously, which may be more robust but less sensitive. For cycle segmentation, besides using midpoints, peaks, and valleys, other identifiable markers can also be used. For warping function estimation, there are also other options [30]–[33]. However, we expect that Kurtek's method [26] used in this paper most likely has the best performance, as it has a very similar signal generation assumption compared to this setting.

3) *Applications*: The primary application of the proposed denoising pipeline is for preprocessing APST data which has low SNR due to physical or equipment limitations, while preserving spatial and temporal resolutions. For artificially stimulated spatiotemporal periodic signals, the proposed method is also useful if cycles need to be aligned.

Due to denoising, cycle averaging improves visualization and detection of minute electrical phenomena with reoccurring patterns. For instance: in hearts with ectopic beats, cycle averaging of exclusive errant beats may illuminate ectopic foci that are not apparent in individual recordings; in the presence of a fixed ratio second-degree atrioventricular block, cycle averaging could be applied to each component of the reoccurring patterns independently, which may delineate a detailed progression of electrical events preceding missing beats. Approaches for dealing with abnormal hearts are further discussed in Section VI-B4.

Besides denoising, it also has other potential applications. For example, the acquired warping functions can be used to classify abnormal and normal cardiac cycles. If some region does not satisfy Assumption 2 due to tissue malfunction, it can be detected by examining the smoothness and standard deviation of the averaged cycle. They can also be used to compare APST data from different hearts, with different diseases, or under different experimental conditions.

4) *Dealing With Cardiac Data Variation*: Real cardiac optical mapping data has variations under different conditions. The proposed pipeline is able to deal with some of these variations. The flexibility mainly comes from the fact that the pipeline allows the cycle averaging to be performed on only a subset of the segmented cycles, which can be detected either manually or automatically depending on the specific situation. We list several potential situations below.

If the data have large cycle duration variance and physiologists consider averaging over all cycles improper, these cycles can be classified into groups according to their cycle durations. Cycle averaging will be done only within each group.

If the data have several ectopic beats, physiologists can distinguish them from normal beats and do averaging separately.

If the data are collected under alternans condition [34] (the heart beats cyclically with alternating strong and weak beats), the original cycle segmentation parameters might not be robust

enough to capture the right cycle separator. In this case, users can make corrections to specify the cycles to be averaged.

If heart rate is very high, atrium depolarization and ventricle repolarization might overlap. In this case, there is no common resting period shared by atrium and ventricle, which means the cycle separators cannot be in the resting period for all signals. To deal with this, there are several options based on specific application scenarios: if depolarization is the major concern, cycle breaks can be chosen somewhere during ventricular repolarization; if repolarization is the major concern, cycle breaks can be chosen somewhere during atrial depolarization but after ventricular repolarization; if the analyses on atrium and ventricle can be separated, atrial signals and ventricular signals can have their own parameters.

Although we cannot cover all physiological situations, readers have the freedom to impose criteria based on their specific conditions.

5) *Limitations*: The capabilities of the proposed method is limited by the signal generation assumptions. In Assumption 1, we do not consider signal drift, vertical scaling, translation and other potential deformations of the signal. Also, we do not consider data beyond Assumption 2, which means signals from different locations that do not share the same cycle segmentation and warping parameters. Any of these factors may introduce errors.

When the number of cycles is relatively small, the SNR gain increases linearly with respect to the logarithm of the number of cycles. This means in order to keep the SNR gain increasing linearly, we need exponentially many cycles, which is impractical. Acquiring a large number of cycles requires long recording periods, during which the heart might move or even change its physiological properties. These undesired effects fail Assumption 1, which assumes different cycles are randomly generated from the same unknown deterministic signal. This could explain the slight drop of SNR gain and SNR when we increased the number of cycles from 160 to 320 [shown in Fig. 6(c) and (d)], as it doubled the recording time.

Additionally, our current programs cannot run in real time yet. The run-time also increases at a empirically quadratic speed with respect to frames per cycle.

6) *Future Work*: As the first attempt on cycle segmentation and alignment for denoising APST data, the methods introduced in this paper likely have many possible future extensions and applications.

The proposed algorithm has been found to provide excellent performance on our real ZHOM data according to visual assessment and brief quantitative evaluation. To understand the performance of the proposed method thoroughly, it is important to do a quantitative analysis on properly generated phantom data. This is because phantom data provides ground truth signals that can be compared with the estimated results. Quantitative analyses should include testing failure conditions, relationship between warping type and SNR gain, relationship between the number of cycles and SNR gain, and performance variation given any change to components of the pipeline.

The proposed model can also be improved by incorporating more types of signal deformation. Also, incorporation of domain knowledge of specific data will likely help the overall al-

gorithm especially in the reference image selection and cycle segmentation stages.

In the proposed method, we used the IDS to estimate warping. Alternatively, one can also use all the signals to estimate the warping simultaneously, by modifying the distance measure in Kurtek's method [26]. However, the run-time would be a bigger issue in that case.

Finally, it is interesting to investigate approaches that use fewer cycles to acquire a higher SNR gain, and optimize current programs and methods to reduce run-time.

APPENDIX

PROPERTIES OF IMAGE DIFFERENCE SIGNAL

We consider the properties of individual cycles of image difference signal $\rho_i(t)$'s

$$\begin{aligned}\rho_i(t) &= \sum_{x,y} (I_t(x,y) - I_{\text{ref}}(x,y))^2 \\ &= \sum_{x,y} (f_i^{x,y}(t) - I_{\text{ref}}(x,y))^2 \\ &= \sum_{x,y} (g^{x,y}(\gamma_i(a_i t)) + \varepsilon_i^{x,y}(t) - I_{\text{ref}}(x,y))^2 \\ &= \sum_{x,y} (\varepsilon_i^{x,y}(t) + d_i^{x,y}(t))^2 \\ &= \sigma^2 \sum_{x,y} \frac{(\varepsilon_i^{x,y}(t) + d_i^{x,y}(t))^2}{\sigma^2}\end{aligned}$$

where σ^2 is the variance of $\varepsilon_i^{x,y}(t)$ and $d_i^{x,y}(t)$ represents $g^{x,y}(\gamma_i(a_i t)) - I_{\text{ref}}(x,y)$. The term $\sum_{x,y} (\varepsilon_i^{x,y}(t) + d_i^{x,y}(t))^2 / \sigma^2$ follows a non-central chi-squared distribution [35], with $k = \sum_{x,y} 1$ and $\lambda = \sum_{x,y} d_i^{x,y}(t) / \sigma^2$. Therefore, we have the expectation and variance of $\rho_i(t)$

$$\begin{aligned}E(\rho_i(t)) &= \sum_{x,y} \sigma^2 + \sum_{x,y} d_i^{x,y}(t)^2, \\ \text{Var}(\rho_i(t)) &= 2 \sum_{x,y} \sigma^4 + 4 \sum_{x,y} d_i^{x,y}(t)^2 \sigma^2.\end{aligned}$$

Then, we separate $\rho_i(t)$ into two parts

$$\rho_i(t) = \eta(\gamma_i(a_i t)) + \mathcal{E}_i(t),$$

where

$$\begin{aligned}\eta(t) &= E(\rho_i(t)) \\ &= \sum_{x,y} ((g^{x,y}(t) - I_{\text{ref}}(x,y))^2 + \sigma^2), \\ \mathcal{E}_i(t) &= \rho_i(t) - E(\rho_i(t)).\end{aligned}$$

Here, $\eta(t)$ acts as the original single cycle signal for the entire IDS $\rho(t)$; different $\rho_i(t)$'s are all considered as random samples generated from $\eta(t)$. $\mathcal{E}_i(t)$ is a zero mean error term with $\text{Var}(\mathcal{E}_i(t)) = 2 \sum_{x,y} \sigma^4 + 4 \sum_{x,y} (g^{x,y}(\gamma_i(a_i t)) - I_{\text{ref}}(x,y))^2 \sigma^2$.

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